

StepSetGo:
Making fitness
fun and
accessible to
everyone by
rewarding
every step they
take

Tell us your challenge. We're here to help.

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[About StepSetGo](#)



StepSetGo encourages people to stay fit by rewarding each step. With the support of Google Cloud Kubernetes Engine, we reduce infrastructure burden and scale quickly.

About StepSetGo

StepSetGo is a mobile fitness reward platform with a goal to make fitness a fun and rewarding journey for everyone by allowing users to convert their steps into credit so that they can redeem lifestyle rewards. Launched in 2019, the startup currently has more than 7 million users.

Industries: Technology

Location: India

Google Cloud results

- Reduces infrastructure management time by more than 40% so DevOps can focus on tasks related to improving overall productivity of the team
- Uses data from BigQuery to gain deeper insights on user behavior to improve app features
- Ensures zero downtime by directing traffic to the appropriate channels with Cloud Load Balancing
- Enables data analytics to help marketing and sales

Support
7M+
enthusiasts
with
requirements



About Searce

Searce is a Google Cloud Premier Partner on a mission to futurify businesses by leveraging cloud, automation, and analytics. It helps businesses through digital transformation by replacing on-premises applications with cloud solutions.

ABOUT Searce

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

solutions.

teams improve
brand
communication
to grow user
base

As people in India become increasingly health conscious, they're turning to home workout videos, live online sessions, and recorded fitness regimes to help them keep fit. But without a source of motivation, it can be difficult to stay committed to regular exercise routines. Tapping into this growing desire to stay fit and to help people achieve their fitness goals, StepSetGo (<https://www.stepsetgo.com/>) makes fitness an accessible and fun activity for everyone. The concept is simple. It rewards people for walking.

The founders of StepSetGo noticed that their friends and family would sign up for gym memberships or make fitness goals at the start of the year, but quickly run out of steam by February or March. This observation inspired them to create something that could help keep people motivated to stay fit throughout the year. With a points system that rewards people for simply walking, the company hopes to encourage an active lifestyle. It does this by letting users redeem lifestyle products that range from electronics to apparel from the app's marketplace using their points.

A cloud-native startup, StepSetGo has garnered more than seven million users in less than two years. To support this rapid growth and scale to

serve even more users, the startup decided to migrate to Google Cloud (<https://cloud.google.com/>) in late 2019. Anyone with a smartphone can access the app for free and join their friends in their fitness journey. It relies on the scalability of Google Kubernetes Engine (/kubernetes-engine) to manage the thousands of requests it gets each second and BigQuery (/bigquery) to run analytics that helps the app continually improve its service. It also uses Cloud Load Balancing (/load-balancing) to ensure that its applications run smoothly without downtime.

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*—Abhay Pai, CTO and co-
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Keeping up with a fast-growing user base

Knowing that there is already a lot of competition for fitness apps in India, the StepSetGo team realized that one way it could stand out was to ensure a seamless and smooth user experience. The user interface (UI) needed to be simple and easily understandable so that there would not be drop-offs once users signed up. "We wanted to capture the market and stand out from the crowd by being an intuitive and reliable app, so the user experience had to be flawless," says StepSetGo CTO and co-founder of Abhay Pai.

Abhay and his team always knew they had to containerize the application in order to avoid bottlenecks as the popularity of the app grew. However, what they did not anticipate was the speed of growth. When it launched, the app had about 50 to 100 users joining the platform in a day. However, that number quickly grew to 56,000 in under three months, and the team found themselves manually

scaling the servers up and down to meet demand. StepSetGo was initially built on a single virtual machine (VM), running on one CPU, but because of the exponential spike in traffic, it now needed to run up to three VMs at once, with its operations team manually updating servers to make sure that the systems were stable. This became a daily routine, taking away time that could have been better spent enhancing the app.

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"Our priority was to stabilize the front end to improve user experience, but we were also beginning to see a growth in traffic, so it was crucial that we scale our back end quickly. At one point, we saw roughly 8,000 requests per second! That's when we decided to migrate to Google Cloud," says Abhay.

StepSetGo had also experienced any startup's biggest nightmare premigration, when its entire infrastructure went down for a day. "This was unacceptable and many of our users were not happy," says Abhay. Since the migration, the team has experienced zero downtime around scale related issues, despite not having to worry about infrastructure.

"With Google Kubernetes Engine and Cloud Load Balancing, we're now able to manage traffic flow automatically without worrying about the infrastructure. This has freed a lot of time for our engineers so that we can focus on developing new and exciting features for the app," he says. Examples include enhancing social interactions in the Friends section where people can connect with each other, and building Turbo Time, which is a feature that

reminds users to walk every once in a while, or to increase their step count and rewards, and finally, more customized challenges that go beyond users' social circle to include companies and corporate teams. While engineers used to spend 50% of their time managing the infrastructure, that has been reduced to less than 10% of their daily working hours after the migration.

A seamless migration through carefully planned CI/CD pipelines

StepSetGo partnered with Google Cloud Premier Partner Searce (<https://www.searce.com/home>) to migrate from its previous cloud provider to Google Cloud. "Being first-time entrepreneurs ourselves, it was challenging to run an entire application from the ground up, from building features to managing the backend infrastructure. Having a team of experts from Searce helped ease this process."

The company built new CI/CD pipelines to move its services to Google Cloud, while building a robust microservices infrastructure. It separated each feature so that if one of its services had a problem, the other features would not be affected. For example, if the step-counting-to-points-conversion API was having issues, it would not affect the bazaar, which is the section on the app where people can redeem points for rewards. Cloud Load Balancing then helps to distribute the different incoming requests by connecting them to the right service.

"We presented the entire architecture of our backend system and had multiple discussions with Searce on the best possible execution. The Searce team then came up with a migration strategy, and after we added our thoughts, we were able to migrate the entire infrastructure onto Google Kubernetes Engine and all of our data into BigQuery."

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—Abhay Pai, CTO and co-founder, StepSetGo

Gathering user data for deeper insights with BigQuery

With BigQuery as its data warehouse, StepSetGo has a better understanding of its user base as it can run quick analyses around specific data points. For example, by pulling data around the time of day when users are most active, it can create new features like challenges where people can connect with each other and compete on steps taken or distance walked. These features enhance the gamification element of the app and motivate people to keep going.

Another benefit of BigQuery is its ability to generate data that helps the marketing and sales teams understand what exactly users are looking for, so they can tap into those insights to place appropriate and relevant ads to potential customers. For example, StepSetGo could run a query on the most redeemed items in the marketplace and use those rewards to attract new users. It also equips the Product team with details that help them fine-tune new feature rollouts to meet users' needs in the most accurate way possible.

StepSetGo also uses BigQuery to create multiple dashboards specific to the marketing and product teams, so that they are able to quickly access information they need. "By using specific

dashboards on BigQuery, the product team can quickly access relevant data and run analyses to understand exactly what is happening at any time. It takes away the guesswork so we can make improvements to the app faster," says Abhay.



Safeguarding the app and building new features with AI and ML

Moving forward, StepSetGo will continue to explore the capabilities of BigQuery, such as using its AI capabilities to detect fraudulent users who attempt to cheat the system. The app offers a referral bonus, so people may create fake accounts to get extra points. To keep the app genuine, it plans to use BigQuery to detect fake or duplicate accounts. This is done by associating a score to every user who invites their friends or family onto the app. The app is now at an 80% accuracy on fraud detection, so it still runs manual checks even after specific accounts are being flagged. "Of course, our goal is

to use BigQuery to make this a foolproof feature in the near future," says Abhay.

The team is also exploring payment gateways so that users can purchase items offered by StepSetGo's partners on the app marketplace with cash. Currently, they can only use their fitness points. "There are a few things in the pipeline, and we are eager to explore different products that Google Cloud offers. Our experience has been a positive one, and we look forward to improving our app even more, with Searce and Google Cloud," says Abhay.